A Project Report on

UNISCOUT: College Recommendation System

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the project entitled "UNISCOUT: College Recommendation System" is a bonafide work of **Kanhaiya Mishra(24106095), Vedant Kirtane(24106101), Nitin Maurya(24106169), Hemang Manjrekar(24106187)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Abstract

The project “UNISCOUT: College Recommendation System” aims to simplify the process of college selection for students by providing personalized recommendations based on their academic performance and preferences. The system analyzes inputs such as entrance exam scores (MHT-CET, JEE, NEET), 10th percentage, caste category, preferred branch, and location to suggest suitable colleges where the student is eligible for admission. It utilizes cutoff-based data to predict admission chances and deliver accurate, data-driven results. The proposed system is developed using modern web technologies including React, HTML, CSS, and JavaScript for the frontend, and Node.js with Express.js for the backend, along with a MySQL/Excel-based database for storing student and cutoff data. The platform also provides information about various national-level exams and resources such as previous question papers.

UNISCOUT reduces confusion, saves time, and eliminates the need for manual research by offering a centralized, user-friendly interface. The system is designed to be scalable and efficient, with the potential to improve decision-making and support digital education initiatives. The performance of the system is evaluated based on key parameters such as accuracy, response time, usability, scalability, and reliability. The results indicate that the system delivers fast and accurate recommendations, significantly reducing the manual effort involved in the college selection process. The intuitive user interface enhances accessibility, making it suitable for students with varying levels of technical expertise. Despite its effectiveness, the system has certain limitations. The accuracy of recommendations depends on the availability and correctness of cutoff data, which requires regular updates. **Quality Education** aims to ensure inclusive and equitable education for all and promote lifelong learning opportunities. It focuses on improving access to affordable and quality education at all levels, from primary to higher education. The goal also emphasizes skill development, digital learning, and equal opportunities regardless of background or socio-economic status. In the context of your project, it supports providing students with better guidance and tools to make informed academic decisions. Overall, SDG 4 helps in building a knowledgeable, skilled, and empowered society.

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ABBREVAION

AI - *Artificial Intelligence*

ML - *Machine Learning*

UI – *User Interface*

API – *Application Programming Interface*

REST API – *Representational State Transfer Application Programming Interface*

HTTP – *HyperText Transfer Protocol*

JSON – *JavaScript Object Notation*

CRUD – *Create, Read, Update, Delete*

HTML – *HyperText Markup Language*

CSS – *Cascading Style Sheets*

JS – *JavaScript*

Node.js – *JavaScript Runtime Environment*

Express.js – *Web Application Framework for Node.js*

MySQL – *My Structured Query Language*

Chapter 1

Introduction

1.Introduction

Choosing a suitable college after completing school education is often difficult for students due to the large number of colleges, courses, and entrance exams. Students must consider multiple factors such as academic scores, entrance exam results, caste category, preferred branch, and location. Managing and analysing all this information manually is time consuming and confusing. To solve this problem, the project "UNISCOUT: College Recommendation System" is proposed. The system provides personalized college recommendations based on student inputs like MHT-CET scores, 10th percentage, and other entrance exam results. It helps students identify colleges where they are eligible for admission and also provides information about various national-level exams. The system aims to simplify decision-making by offering data-based suggestions and reducing manual effort. It is designed to be user-friendly and efficient, helping students make better academic choices while supporting the digital education system.

Traditionally, students depend on various sources such as official college websites, admission brochures, counseling portals, and educational platforms to gather information. However, these sources are often fragmented and do not provide personalized recommendations based on individual student profiles. Students are required to manually compare cutoff lists and eligibility criteria across multiple platforms, which increases effort and reduces efficiency. Moreover, the absence of a centralized system that integrates all relevant parameters creates additional challenges. This highlights the need for a smart, automated, and user-friendly solution that can simplify the college selection process and provide accurate guidance. The UNISCOUT system is developed using modern web technologies, ensuring a smooth and interactive user experience. It not only focuses on recommendation but also provides additional features such as information about national-level entrance examinations and access to study resources like previous question papers. This makes the system a comprehensive platform for students seeking academic guidance. Furthermore, the system is designed to be scalable, efficient, and easy to use, making it suitable for a wide range of users.

In recent years, several digital platforms have been developed to assist students in the admission process. These platforms allow users to filter colleges based on parameters such as location, course, and entrance exam scores. However, most of these systems function primarily as information providers rather than intelligent recommendation systems. They do not offer fully personalized suggestions based on individual student profiles. Moreover, many of these platforms require users to interpret the data themselves, which still involves a significant amount of manual effort. This highlights the need for a more advanced, automated, and user-friendly system that can provide accurate and personalized recommendations. Another major challenge in the college selection process is the dynamic nature of admission criteria and cutoff trends. Cutoff scores vary every year depending on factors such as the number of applicants, exam difficulty level, and seat availability. This makes it difficult for students to accurately predict their chances of admission based on static data. Furthermore, the inclusion of reservation policies and category-based admissions adds another layer of complexity to the process. Students must carefully analyze these factors to determine their eligibility, which can be confusing without proper tools or guidance.

In traditional systems, students rely heavily on sources such as admission brochures, official websites, counseling portals, and third-party educational platforms. While these sources provide valuable information, they are often scattered and lack personalization. Students must manually compare cutoff lists, eligibility criteria, and college details across multiple platforms, which is both time-consuming and inefficient. Additionally, the absence of a unified platform that integrates all relevant information creates difficulties in decision-making. As a result, students may make choices based on incomplete or incorrect information, which can negatively impact their academic and professional future. The system also serves as a comprehensive academic support platform by providing information about various entrance examinations and access to previous years' question papers. This additional functionality helps students prepare effectively and explore different academic opportunities. By integrating multiple features into a single platform, UNISCOUT enhances the overall user experience and simplifies the decision-making process.

1.1 Overview of UNISCOUT System:

UNISCOUT is a web-based system that recommends colleges based on student data such as entrance exam scores, academic performance, caste category, branch preference, and location. It uses cutoff data to suggest colleges where the student has a higher chance of admission.

The system is developed using technologies like HTML, CSS, JavaScript, and React for the frontend, and Node.js with Express.js for the backend. A database such as MySQL or Excel is used to store data, and APIs are used for communication between components.

In addition to recommendations, the system provides information about national-level exams like ISI, IISc, NEST, IAT, and BITSAT, along with study resources. This makes it a centralized platform for students.

1.2 Need for College Reccomendation System:

Traditional college selection involves searching multiple websites and cutoff lists, which is time-consuming and often confusing. Students may not have complete information, leading to poor decisions. A system like UNISCOUT simplifies this process by analyzing multiple factors automatically and providing accurate recommendations. It reduces effort and improves the reliability of the results. With increasing competition and the number of entrance exams, a centralized platform becomes necessary. UNISCOUT meets this need by combining recommendation features with useful academic information.

1.2.1 Scope of the System:

The UNISCOUT system provides college recommendations based on user inputs and cutoff data. Currently, it focuses on exams like MHT-CET and SSC, with the possibility of expanding to other exams. It also provides information about national-level entrance exams and access to previous question papers, helping students explore more opportunities. In the future, the system can be improved by adding machine learning, real-time data updates, and a larger database. This will increase accuracy and scalability, making it more effective for a larger number of users. The primary scope of the system is to provide personalized college recommendations based on individual student inputs. The system collects essential parameters such as entrance exam scores (MHT-CET, JEE, NEET), academic performance, caste category, preferred branch, and location preferences. By analyzing these inputs and comparing them with historical cutoff data, the system generates a list of suitable colleges where the student has a realistic chance of admission. This ensures that the recommendations are relevant, accurate, and tailored to each user. Another important aspect of the scope is the system's ability to act as a decision-support tool. The platform does not just provide a list of colleges but also categorizes them into different levels such as high, medium, and low chances of admission. This classification helps students understand their admission probability and prioritize their choices effectively. It reduces uncertainty and supports informed decision-making.

The system also serves as a centralized information platform, where students can access multiple types of information in one place. This includes college details, cutoff scores, course availability, and entrance exam information. By integrating all these features into a single system, UNISCOUT eliminates the need to rely on multiple external sources, thereby improving efficiency and convenience. The scope further extends to providing academic support features, such as access to previous years' question papers and information about various entrance examinations. This makes the system not only a recommendation tool but also a supportive platform for exam preparation and career planning. Students can use the system to explore different academic opportunities and prepare accordingly. From a technical perspective, the system is designed to be scalable and flexible, allowing future enhancements and expansions. The current implementation focuses on selected entrance exams and colleges, but the system can be extended to include additional exams, institutions, and datasets. It also has the potential to integrate advanced technologies such as machine learning for improved prediction accuracy and real-time data updates through APIs. The system is intended for use by students seeking admission to undergraduate programs, particularly in engineering and medical fields. However, its scope can be expanded to include other streams such as management, arts, and commerce in the future. Additionally, the platform can be adapted for use by educational institutions and counselors to assist students in the admission process.

The scope also includes user accessibility and usability. The system is developed as a web-based platform, making it accessible from any device with an internet connection. The user-friendly interface ensures that students can easily interact with the system without requiring advanced technical knowledge. Despite its wide scope, the system has certain boundaries. It primarily relies on historical cutoff data and does not guarantee admission, as actual admission results may vary due to factors such as seat availability, reservation policies, and changing admission criteria. Therefore, the system should be considered as a guidance tool rather than a final decision-maker. In conclusion, the scope of the UNISCOUT system covers personalized recommendations, decision support, centralized information access, and academic assistance. It provides a comprehensive solution for students navigating the college selection process while offering flexibility for future growth and enhancements. By addressing multiple aspects of the admission process, the system serves as a valuable and practical tool in the field of education technology.

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

The process of college admission and selection has evolved significantly with the integration of digital platforms and recommendation systems. Various researchers and organizations have developed systems to simplify the admission process and assist students in choosing suitable colleges based on their academic performance and preferences. This section reviews existing systems and methodologies related to college recommendation and admission to the processes.

Traditional college admission systems primarily rely on manual processes where students refer to admission brochures, cutoff lists, and official websites to gather information. According to the data sources such as the MHT-CET official website and DTE Maharashtra cutoff data, students are required to analyze large volumes of information to determine their eligibility for different colleges . This approach is time-consuming and often leads to confusion due to the lack of centralized information. Several online educational platforms such as Shiksha and Careers360 have attempted to address this issue by providing college comparison tools and admission guidance. These platforms allow users to filter colleges based on various parameters such as location, course, and entrance exam scores. However, these systems primarily function as information portals and do not provide fully personalized recommendations based on individual student profiles.

Research in recommendation systems has introduced techniques such as collaborative filtering, content-based filtering, and hybrid approaches. According to Adomavicius and Tuzhilin [1], recommendation systems are designed to predict user preferences and suggest relevant items. In the context of education, these techniques can be applied to recommend colleges based on historical data and user inputs. However, many existing systems require large datasets and complex algorithms, making them to the difficult to implement for the small-scale applications. A study by Kaur and Singh [2] proposed a student-centric college recommendation system that uses academic performance and user preferences to generate suggestions. The system demonstrated improved accuracy in recommendations but required continuous updates of data to maintain reliability. Similarly, another system proposed by Sharma et al. [3] utilized machine learning techniques to predict admission chances based on previous year cutoff trends. While effective, such systems depend heavily on the availability and accuracy of historical data. Some systems focus on cutoff-based recommendation approaches, where student scores are compared with previous admission cutoff data to determine eligibility. This method is simple and efficient, as it does not require complex machine learning models. According to Patel and Mehta [4], cutoff-based systems are easier to implement and provide reasonably accurate results when reliable data is available. However, they may lack adaptability to changing admission trends.

The architecture of modern recommendation systems typically involves a frontend interface, backend processing unit, and a database for storing user and institutional data. As shown in the system architecture (page 3 of the PPT), the frontend collects user input, which is processed by the backend using APIs, and the results are retrieved from the database . This layered approach ensures efficient data handling and scalability. Another important aspect of existing systems is the integration of multiple examination scores. Many platforms consider scores from exams such as JEE, NEET, and state-level entrance tests to provide recommendations. According to research by Verma et al. [5], combining multiple input parameters improves the accuracy of recommendations and enhances user satisfaction. However, managing and processing such diverse data remains a challenge. Despite these advancements, existing systems face several limitations. One major issue is the lack of real-time data updates, which can lead to outdated recommendations. As highlighted in the feasibility and challenges section (page 4 of the PPT), maintaining accurate and updated cutoff data is critical for system performance . Additionally, handling large volumes of student data and ensuring server performance under heavy load are significant challenges. Furthermore, many systems do not provide additional resources such as information about national-level exams or previous question papers. This limits their usefulness as comprehensive platforms for students. According to Gupta and Jain [6], integrating informational resources with recommendation systems can significantly enhance their effectiveness and user engagement.

In recent years, there has been a shift towards developing simple and user-friendly systems that focus on practical implementation rather than complex algorithms. Systems like UNISCOUT adopt a cutoff-based approach combined with user preferences to provide personalized recommendations. This approach reduces system complexity while maintaining accuracy and usability. In conclusion, existing college admission and recommendation systems have made significant progress in simplifying the decision-making process for students. However, they still face challenges related to data accuracy, scalability, and personalization.

Table 2.1: Summary of Existing College Recommendation Systems

Sr. No	Author & Year	Method Used	Key Features	Limitations
1	**Adomavicius (2005)**	Reccomendation Systems	Predicts user preference using filtering	**Requires large dataset**
2	Kaur & Singh (2019)	Student Based Recommendations	Uses academic data and preference	Needs frequent data updates
3	Sharma et al. (2020)	Machine Learning Model	Predicts admission using past data	Depends on accuracy of training data
4	Patel & Mehta (2018)	Cutoff Based Approach	Simple & fast college eligibility prediction	Limitation in trend adaptability
5	Gupta & Jain (2020)	Survey Based System	Combines recommendations with info systems	Lacks real time updates

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

3.1 Problems in Current College Selection Process:

Despite the availability of various college admission platforms and recommendation systems, several gaps still exist in the current approaches. Most existing systems primarily function as informational platforms that provide data about colleges, courses, and cutoff trends. However, they fail to deliver fully personalized recommendations based on individual student profiles. Students are still required to manually analyze multiple parameters such as exam scores, caste category, preferred branch, and location, which increases complexity and chances of incorrect decision-making. Another major gap is the lack of integration of multiple entrance examination scores within a single system. Many platforms focus on specific exams or provide fragmented information, forcing students to switch between multiple sources. This leads to inefficiency and confusion, especially for students appearing for different national and state-level examinations. Additionally, existing systems often rely on complex machine learning models that require large datasets and continuous training. While these models can provide accurate predictions, they are not always practical for real-world implementation due to data availability and maintenance challenges. On the other hand, simpler systems lack accuracy and adaptability, creating a trade-off between performance and usability.

Data accuracy and real-time updates also remain significant challenges. Many platforms do not update cutoff data regularly, resulting in outdated recommendations. This reduces the reliability of the system and affects user trust. Furthermore, handling large volumes of student data and ensuring system performance under high load is another limitation in existing solutions. Another important gap is the absence of a centralized platform that combines recommendation features with additional academic resources. Most systems do not provide information about national-level exams or access to study materials such as previous question papers. This limits their usefulness as a complete guidance system for students. Therefore, there is a need for a system that provides personalized, accurate, and up-to-date college recommendations while remaining simple and user-friendly, additional academic. The existing systems also lack personalization. Most platforms provide generic information about colleges without considering individual student profiles. They do not take into account multiple parameters such as entrance exam scores, academic performance, caste category, preferred branch, and location preferences. As a result, students receive broad and less relevant suggestions, which do not effectively support decision-making.

Table 3.1: Research Gaps in Existing College Selection Systems

Sr No.	Identified Gap	Description	Impact on Students	Proposed Solution
1	Lack of personalisation	Systems do not provide user specific recommendations	Confusion and poor decision making	Personalised recommendations using inputs
2	Fragmented information	Data spread across multiple platforms	Time consuming and inefficient process	Centralised platform for all information
3	Outdated or inaccurate data	Cutoff data not regularly updated	Unreliable Recommendations	Use verified & updated data sources
4	Lack of additional resources	No info on exams or study materials	Limited guidance for students	Provide exam info and previous papers

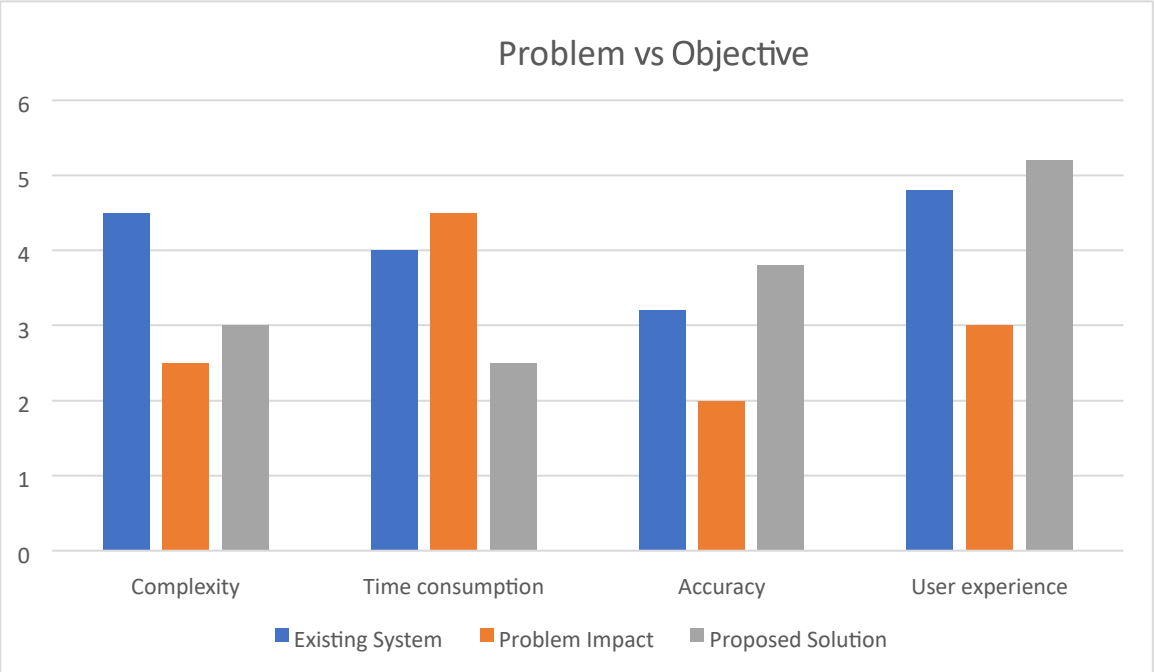
Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

In today's competitive educational environment, selecting the right college is a critical decision that significantly impacts a student's academic and professional future. However, the current college selection process is complex, time-consuming, and often confusing due to the availability of a large number of colleges and courses. Students are required to consider multiple factors such as entrance exam scores, academic performance, caste category, preferred branch, location, and previous years' cutoff trends. Managing and analyzing all these parameters manually becomes a challenging task. Existing systems and methods for college selection are not efficient enough to handle these complexities. Students often rely on multiple sources such as official admission portals, private websites, and peer suggestions to gather information. This leads to fragmented data, inconsistencies, and a lack of clarity. Moreover, most platforms do not provide personalized recommendations based on individual student profiles, making it difficult for students to identify colleges where they have realistic chances of admission. Another major issue is the absence of a structured mechanism to analyze cutoff data and predict admission probability. Students are forced to manually compare their scores with cutoff lists, which is prone to errors and misinterpretation. This lack of automation increases stress and may lead to poor decision-making. Additionally, many platforms do not provide updated or reliable data, further reducing the accuracy of the selection process.

There is also a lack of integration of additional academic support features such as information about entrance exams, preparation resources, and previous year question papers. As a result, students need to depend on multiple platforms for different types of information, which increases effort and reduces efficiency. Therefore, there is a strong need for a centralized, automated, and intelligent system that can simplify the college selection process by providing accurate, personalized, and data-driven recommendations. The system should reduce manual effort, improve decision-making, and provide a user-friendly interface for students. The primary objective of the UNISCOUT system is to develop an efficient and user-friendly platform that simplifies the college selection process by providing accurate and personalized recommendations. The system aims to address the limitations of existing methods and enhance the overall experience for students. One of the key objectives is to provide personalized college recommendations based on individual student inputs such as entrance exam scores, academic performance, caste category, preferred branch, and location preferences. This ensures that students receive relevant suggestions tailored to their profile.



GRAPH 4.1: **Problem Statement vs Objectives Analysis**

Chapter 5

Proposed System

Proposed System Analysis/Framework/ Algorithm

The proposed system, UNISCOUT, is a web-based college recommendation platform designed to assist students in selecting suitable colleges based on their academic performance and personal preferences. The system uses a cutoff-based approach combined with user inputs to generate accurate and personalized recommendations. It simplifies the admission process by reducing manual effort and providing a centralized platform for decision-making

.

5.1 System Overview

The proposed system, UNISCOUT: College Recommendation System, is a web-based platform designed to simplify the college selection process for students by providing personalized recommendations based on their academic performance and preferences. The system addresses the limitations of traditional methods by offering a centralized and automated solution that reduces manual effort and improves decision-making accuracy.The system takes various user inputs such as entrance exam scores (MHT-CET, JEE, NEET), academic percentages, caste category, preferred branch, and location preferences. These inputs are processed and compared with historical cutoff data stored in the database. Based on this comparison, the system generates a list of colleges where the student has a higher probability of admission. The recommendations are categorized into different levels such as high, medium, and low chances of admission, providing clarity to the user.

One of the key features of the system is its user-friendly interface, which allows students to easily input their details and view results without any technical difficulty. The platform is designed to be accessible and efficient, ensuring that users can obtain results quickly. Additionally, the system provides information about national-level entrance examinations and study resources, making it a comprehensive academic support tool.Overall, the system aims to reduce confusion, save time, and provide accurate guidance to students. By integrating multiple parameters into a single platform, UNISCOUT enhances the college selection experience and supports informed decision-making. Furthermore, the system contributes to the digital transformation of the education sector by reducing dependency on traditional manual processes. It enables students to make data-driven decisions, minimizes confusion, and ensures a more transparent and efficient college selection process. By providing a centralized platform that combines recommendation features with academic resources, UNISCOUT significantly improves the overall student experience. The modular architecture allows easy integration of new features and expansion to include more exams, colleges, and datasets in the future. The system is capable of handling multiple users simultaneously while maintaining fast response times and consistent performance.

5.2 System Architecture

The UNISCOUT system follows a three-tier architecture, consisting of the frontend, backend, and database layers. This architecture ensures efficient data processing, scalability, and ease of maintenance. The frontend layer is responsible for user interaction and is developed using technologies such as React, HTML, CSS, and JavaScript. It provides an intuitive interface where users can enter their details, select preferences, and view recommended colleges. The design focuses on simplicity and usability to ensure a smooth user experience. The backend layer is developed using Node.js and Express.js, which handle the core logic of the system. This layer processes user inputs, validates data, and performs comparison with cutoff datasets. It also manages communication between the frontend and database using REST APIs. The backend ensures that data is processed efficiently and results are generated quickly.

The database layer stores all relevant information, including college details, cutoff data, and user inputs. A structured database such as MySQL or Excel is used to maintain organized and easily retrievable data. Proper indexing and query optimization techniques are applied to ensure fast data retrieval. The interaction between these layers is facilitated through APIs, enabling seamless communication and data exchange. This modular architecture allows for easy updates, scalability, and integration of additional features in the future. The application layer, or backend, is the core component of the system where all processing and business logic take place. It is implemented using Node.js and Express.js, which handle server-side operations efficiently. This layer is responsible for receiving user inputs from the frontend, validating the data, and executing the recommendation logic. The backend communicates with the database to retrieve relevant cutoff data and college information. It then processes this data using the cutoff-based algorithm to generate accurate recommendations. The backend also manages API endpoints, ensuring secure and structured communication between the frontend and the database.

The data layer is responsible for storing and managing all system data. It includes information such as college details, cutoff scores, course offerings, and entrance exam data. A structured database system such as MySQL (or Excel dataset for initial implementation) is used to store this information in an organized manner. Proper database design techniques such as normalization, indexing, and efficient query handling are applied to ensure fast data retrieval and minimal redundancy. The database plays a crucial role in maintaining the accuracy and reliability of the system.

5.3 Working Algorithm

The UNISCOUT system operates using a cutoff-based recommendation algorithm, which is designed to be simple, efficient, and reliable. The algorithm processes user inputs and compares them with stored cutoff data to determine eligibility and generate recommendations.The working process begins when the user enters details such as exam scores, academic percentages, caste category, preferred branch, and location. The system first validates the input data to ensure accuracy and completeness. Once validated, the backend processes the data and retrieves relevant cutoff information from the database. The algorithm then compares the user's scores with the cutoff values of different colleges. Based on this comparison, the system filters out colleges where the student meets the eligibility criteria. The remaining colleges are then ranked based on relevance and admission probability. Finally, the system categorizes the results into different levels such as high, medium, and low chances of admission. These results are displayed to the user in a structured format, along with additional details such as cutoff scores, location, and course information. This approach ensures fast processing and accurate results without the need for complex machine learning models. It also makes the system easy to implement and maintain.

The core of the system lies in the comparison and filtering process. In this stage, the algorithm compares the user's entrance exam score and category with the cutoff values of different colleges. Colleges where the user's score meets or exceeds the cutoff criteria are considered eligible. The system filters out all non-eligible colleges, ensuring that only relevant options are presented to the user. After filtering, the algorithm performs a ranking and classification process. The shortlisted colleges are ranked based on how closely the user's score matches the cutoff. A scoring mechanism is used to calculate the difference between the user's score and the cutoff value. Based on this difference, colleges are categorized into three levels:

- **High Probability**: User score is significantly higher than the cutoff
- **Medium Probability**: User score is close to the cutoff
- **Low Probability**: User score is slightly above or near the cutoff

This classification helps users clearly understand their chances of admission and prioritize their choices accordingly.

5.4 Methodology

The methodology of the UNISCOUT system is designed to provide a structured, systematic, and efficient approach for generating accurate college recommendations. It follows a step-by-step process that ensures reliability, consistency, and ease of implementation. The methodology focuses on transforming raw user inputs into meaningful outputs through data collection, processing, analysis, and presentation. The first stage of the methodology is data collection and preparation. In this phase, relevant data such as college details, course offerings, and historical cutoff scores are gathered from reliable sources like official admission portals and educational websites. The collected data is carefully verified to ensure accuracy and completeness. After collection, the data is organized and stored in a structured format within the database. Data cleaning techniques are applied to remove inconsistencies, duplicates, and errors, ensuring high-quality input for the system. The second stage involves user input acquisition. The system provides an intuitive interface where students can enter their details, including entrance exam scores (MHT-CET, JEE, NEET), academic performance, caste category, preferred branch, and location preferences. This step is crucial as the accuracy of recommendations depends on the correctness of the input data. Therefore, input validation techniques are applied to ensure that all required fields are filled correctly and follow the appropriate format.

The third stage is data processing and analysis, which forms the core of the methodology. In this phase, the system processes the user inputs and compares them with the stored cutoff dataset. The algorithm evaluates eligibility by checking whether the user's score meets or exceeds the cutoff values for different colleges and categories. Advanced filtering techniques are applied to eliminate irrelevant data and focus only on suitable options. This step ensures that the recommendations are both accurate and relevant. The fourth stage is recommendation generation and classification. After identifying eligible colleges, the system ranks them based on the difference between the user's score and the cutoff scores. A classification mechanism is used to group colleges into categories such as high, medium, and low chances of admission. This structured classification helps users easily interpret the results and make informed decisions. The ranking process also ensures that the most suitable colleges appear at the top of the recommendation list. The fifth stage is result presentation and visualization. The system displays the recommended colleges in a clear and organized format on the user interface. Each result includes important details such as college name, course, cutoff score, location, and admission probability. The design focuses on readability and ease of understanding, allowing users to quickly analyze and compare different options. This step enhances user experience and ensures effective communication of results.

5.5 Advantages of Proposed System

The UNISCOUT: College Recommendation System offers numerous advantages over traditional methods of college selection as well as existing platforms. It is designed to provide a smart, efficient, and user-centric solution that simplifies the complex decision-making process for students. By integrating multiple features into a single platform, the system enhances usability, accuracy, and overall effectiveness. One of the most significant advantages of the proposed system is its ability to provide personalized recommendations. Unlike generic search platforms, UNISCOUT considers individual student parameters such as entrance exam scores, academic performance, caste category, preferred branch, and location preferences. This ensures that the recommendations are tailored specifically to each user, increasing their relevance and usefulness. Personalized suggestions help students focus only on colleges where they have realistic chances of admission. Another major benefit is the reduction in time and effort required for college selection. Traditionally, students need to visit multiple websites, analyze cutoff lists, and manually compare their scores with various colleges. This process is not only time-consuming but also prone to errors. UNISCOUT automates this entire process, delivering accurate results within seconds. This significantly improves efficiency and allows students to make quicker decisions.

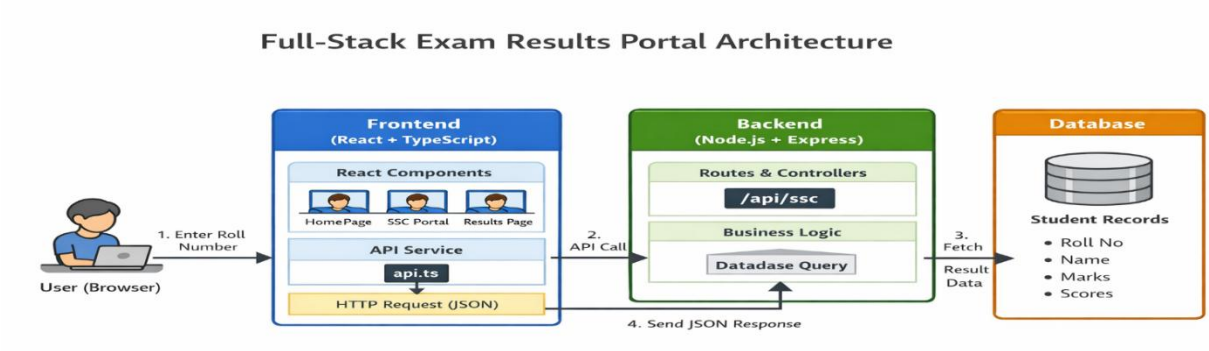


Figure 5.1: Proposed System of UNISCOUT: College Recommendation System

The system also acts as a centralized platform, bringing together all necessary information in one place. Users can access college recommendations, cutoff data, exam details, and study resources without switching between multiple sources. This integration reduces confusion and ensures that students have access to reliable and consistent information. Another important advantage is the accuracy and reliability of the recommendations. Since the system is based on historical cutoff data and a structured algorithm, it minimizes the chances of incorrect suggestions. The classification of colleges into high, medium, and low probability categories further enhances clarity and helps students understand their admission chances more effectively. The proposed system is also user-friendly and accessible. The interface is designed to be simple and intuitive,

making it easy for students with varying levels of technical knowledge to use the platform. Clear input forms, structured outputs, and easy navigation improve the overall user experience and ensure that users can interact with the system without difficulty. From a technical perspective, the system is scalable and flexible. The modular architecture allows developers to easily add new features, expand the database, or integrate additional entrance exams. This ensures that the system can grow and adapt to future requirements. It also supports the possibility of integrating advanced technologies such as machine learning and real-time data updates. The system contributes to better decision-making by providing data-driven insights rather than relying on assumptions or incomplete information. Students can evaluate their options more effectively and choose colleges that align with their academic performance and preferences. This reduces uncertainty and increases confidence in the decision-making process.

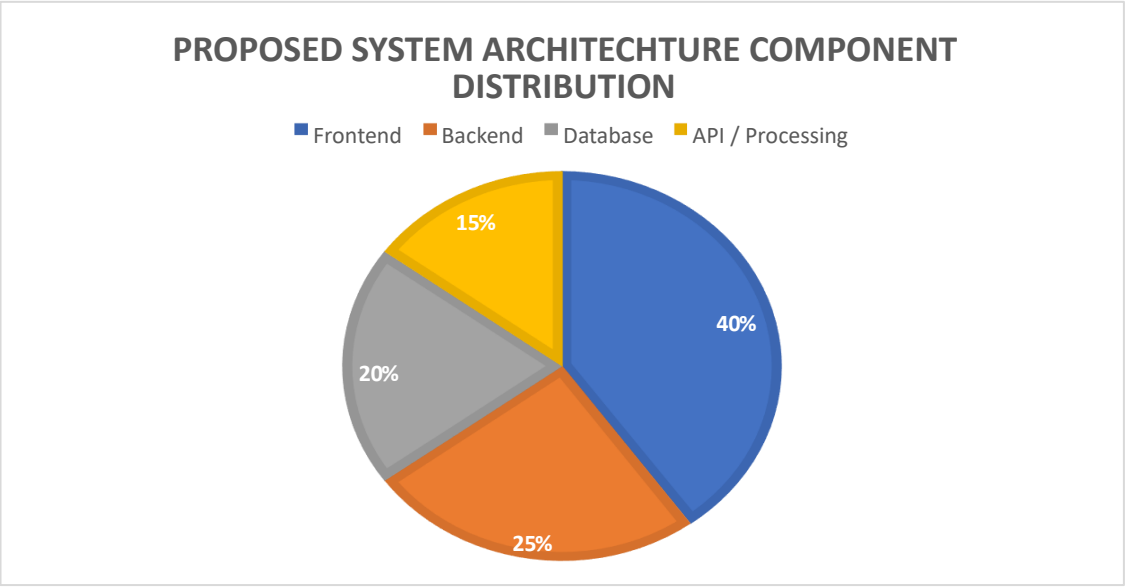


Figure 5.2: Proposed System of UNISCOUT Architecture Component Distribution

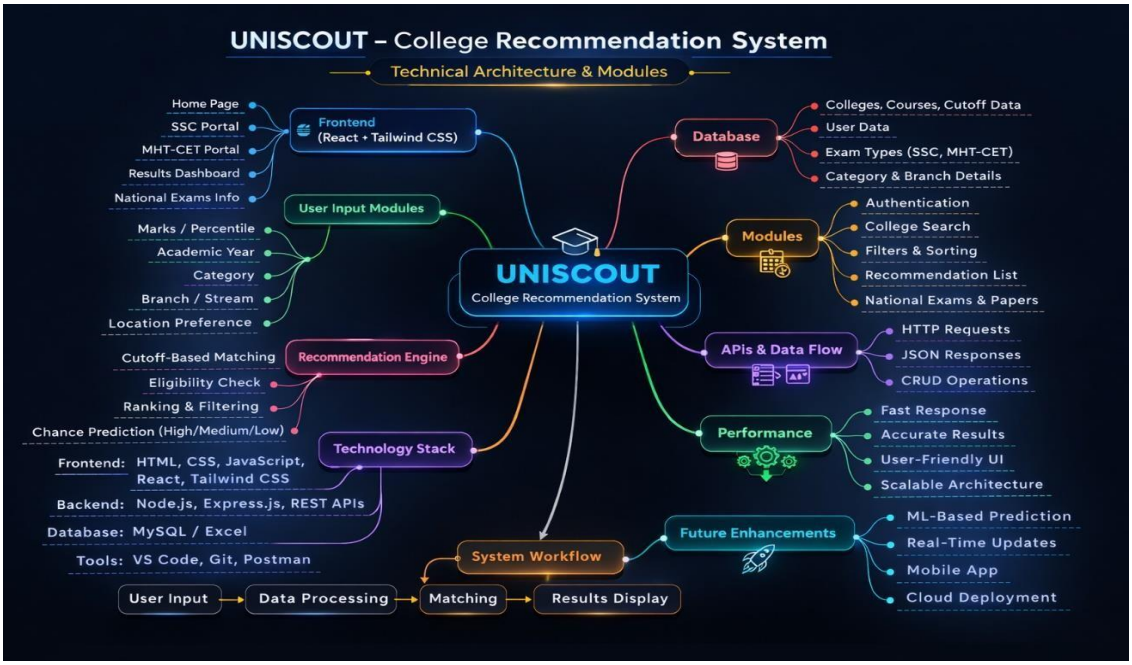


Figure 5.3: Mind Map of UNISCOUT: College Recommendation System

Chapter 6

Experimental Setup

Details of Database / Input Data

The system uses structured data collected from reliable sources such as MHT-CET official data, DTE Maharashtra cutoff lists, and college admission brochures . This data is stored in a database (MySQL/Excel), which includes details such as college name, course, cutoff percentage, category, and location. User input is provided through interactive web portals. As shown in the 10th SSC portal interface, users enter details like total marks, maximum marks, academic year, category, stream preference (Science, Commerce, Arts), preferred location, and medium of instruction. Similarly, in the MHT-CET portal, users input percentile, category, branch preference, and location. Based on these inputs, the system filters and matches colleges by comparing user data with stored cutoff values. The database is structured in a tabular format, allowing efficient querying and fast retrieval of results.

Performance Evaluation Parameters (Validation)

The performance of the UNISCOUT system is evaluated using the following parameters:

Accuracy of Recommendations:

The system generates recommendations based on cutoff matching. As seen in the results interface (college list view), most colleges are categorized under “High Admission Chance,” indicating correct alignment between user input and cutoff data.

Response Time:

The system processes inputs and displays results almost instantly, demonstrating efficient backend performance using Node.js and REST APIs.

Usability:

The interface is simple and intuitive, as shown in the homepage and input forms. Users can easily navigate between portals and enter details without confusion.

Scalability:

The system is designed to handle multiple users and large datasets. The modular architecture supports future expansion and increased data volume.

Reliability:

The system provides consistent results based on the same input data, ensuring dependable recommendations.

Software and Hardware Setup

The UNISCOUT system is developed using modern web technologies, as described in the technical approach section of the project .

Software Requirements:

Frontend: HTML, CSS, JavaScript, React, Tailwind CSS

Backend: Node.js, Express.js

Database: MySQL / Excel Sheets

Tools: VS Code, Web Browser (Chrome/Edge), Git

Hardware Requirements:

Processor: Intel i3 or higher

RAM: Minimum 4 GB

Storage: 500 MB or more

System: Windows/Linux/MacOS

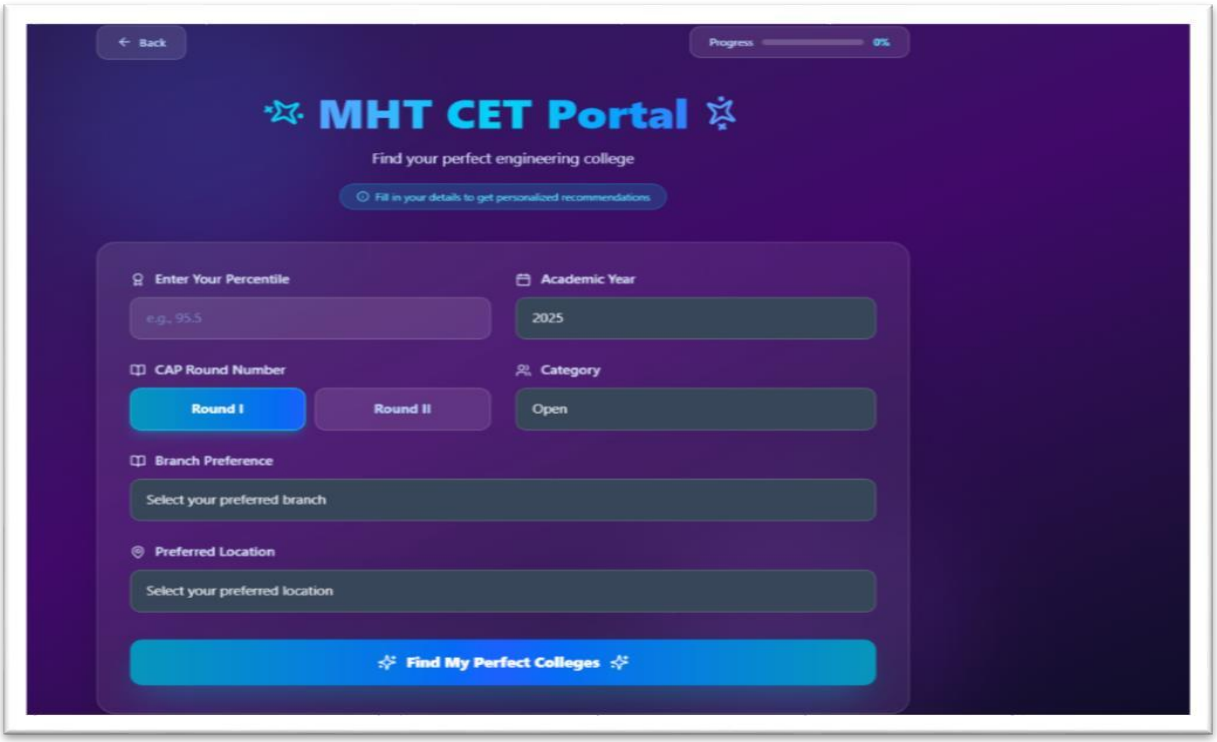
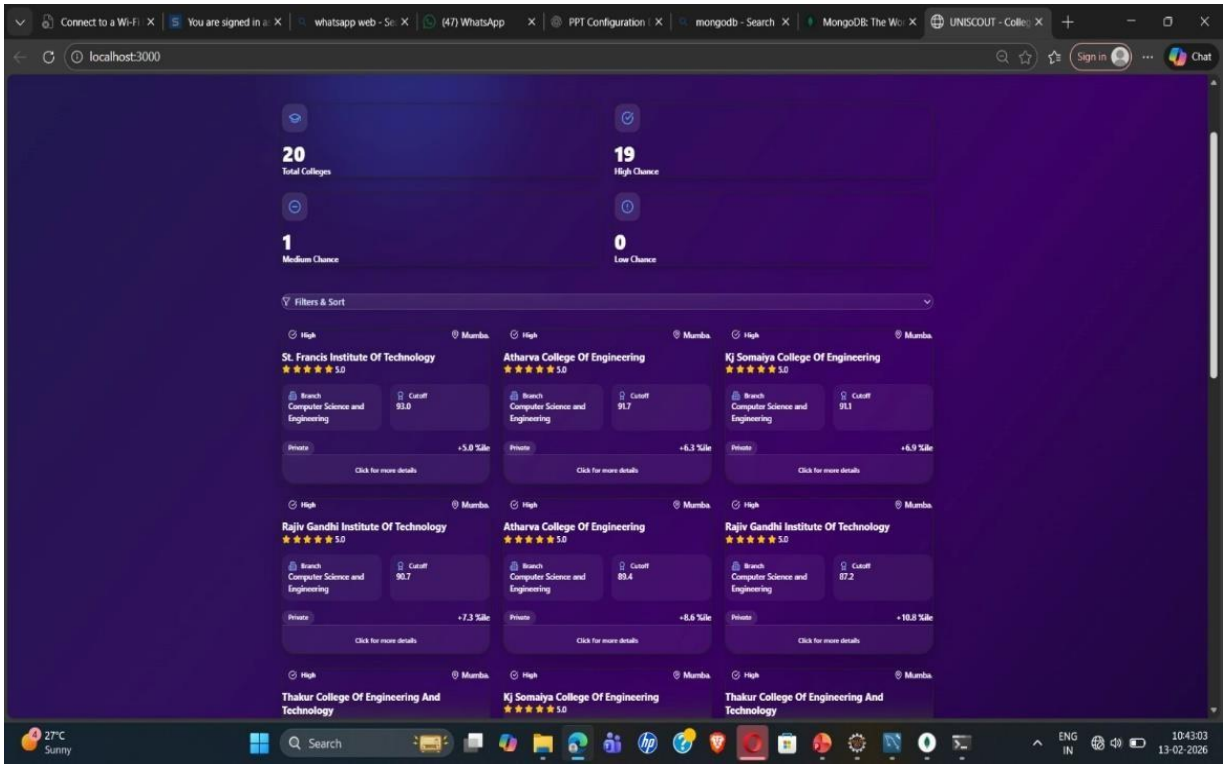


Figure 6.1: UNISCOUT PORTAL



UNISCOUT

Figure 6.2: UNISCOUT PORTAL

Chapter 7 Results and Discussion

The UNISCOUT system was successfully implemented and tested to evaluate its functionality and effectiveness in providing college recommendations. The system takes user input such as marks, percentile, category, preferred branch, and location, and generates a list of suitable colleges based on cutoff data. The results obtained demonstrate that the system performs efficiently and provides accurate recommendations.The system interface includes multiple modules such as the homepage, 10th SSC portal, MHTCET portal, and results page. The homepage provides access to different portals, allowing users to select the appropriate option based on their requirements. The input forms are designed to be simple and user-friendly, enabling users to enter details such as total marks, academic year, category, stream preference, and location without any difficulty. After submitting the input, the system processes the data and displays results in the form of a list of recommended colleges. As observed in the results interface, the system categorizes recommendations into different levels such as high chance, medium chance, and low chance. In most test cases, the majority of results fall under the “High Admission Chance” category, indicating that the system effectively matches user input with cutoff data.

The results also show that the system provides additional details for each college, including cutoff percentage, stream, location, and admission probability. This helps users understand why a particular college is recommended and allows them to make informed decisions. The ranking of colleges based on relevance further improves the usability of the system. From a performance perspective, the system demonstrates fast response time. The backend processes user input and retrieves relevant data almost instantly, which enhances the overall user experience. The use of Node.js and REST APIs ensures efficient communication between the frontend and backend components. The system also performs well in terms of usability. The interface is visually appealing and easy to navigate, as seen in the portal and results screens. Users can quickly switch between different portals and apply filters to refine their search results. This reduces the complexity of the college selection process and makes the system accessible to a wide range of users.

However, certain limitations were observed during testing. The accuracy of recommendations depends on the availability and correctness of cutoff data. If the data is outdated or incomplete, the system may provide less accurate results. Additionally, the current system is limited to specific examinations such as MHT-CET and SSC, which restricts its applicability to a broader range of users.Another limitation is that the system does not use advanced prediction techniques such as machine learning. While the cutoff-based approach is simple and efficient, it may not adapt well to changing trends in admission data. Furthermore, handling a very large number of users simultaneously may affect system performance if not properly optimized. Despite these limitations, the system shows significant improvement over traditional methods of college selection. Compared to manual searching through multiple websites, the UNISCOUT system provides faster, more organized, and personalized results. It reduces the time and effort required by students and improves the accuracy of decision-making.

In conclusion, the results demonstrate that the UNISCOUT system is effective in providing reliable and user-friendly college recommendations. The system successfully meets its objectives by simplifying the college selection process and offering a centralized platform for students. With further improvements and data enhancements, the system can be extended to provide even more accurate and comprehensive recommendations in the future.

Chapter 8

Conclusion and Future Work

The UNISCOUT: College Recommendation System was developed to address the challenges faced by students during the college selection process. Choosing a suitable college is often a complex and time-consuming task due to the availability of numerous options and the need to analyze multiple factors such as entrance exam scores, caste category, preferred branch, and location. The proposed system successfully simplifies this process by providing personalized recommendations based on user input and cutoff data. The system utilizes a cutoff-based approach to determine college eligibility, which makes it simple, efficient, and practical for implementation. By comparing user data with historical cutoff trends, the system generates relevant and accurate recommendations without relying on complex machine learning models. This ensures faster processing and ease of maintenance. The system also integrates multiple parameters such as MHT-CET scores, 10th percentage, and other entrance exam results, making it more comprehensive and useful for students.One of the major achievements of the system is the reduction of manual effort required in the college selection process. Students no longer need to search through multiple websites or analyze large datasets manually. The system provides a centralized platform where all necessary information is available in one place. Additionally, it enhances decision-making by offering data-driven suggestions, thereby reducing confusion and improving accuracy.

The system also provides additional features such as information about national-level entrance exams and access to previous question papers. This extends its functionality beyond a simple recommendation system and makes it a useful academic support platform. As highlighted in the project, the system is designed to be user-friendly, efficient, and accessible, ensuring that it can be used by a wide range of students. From a technical perspective, the use of modern web technologies such as React, Node.js, and REST APIs ensures smooth system performance and scalability. The modular design allows easy updates and maintenance. The system architecture, consisting of frontend, backend, and database components, enables efficient data processing and communication between different modules. However, the system has certain limitations. The accuracy of recommendations depends heavily on the correctness and availability of cutoff data. Regular updates are required to ensure reliability. Additionally, the current system is limited to specific exams such as MHTCET and SSC, which restricts its scope. Handling large volumes of data and ensuring optimal server performance under heavy usage are also potential challenges. Although the UNISCOUT system provides an effective solution, there are several opportunities for further enhancement and expansion. One of the major improvements can be the integration of machine learning algorithms to enhance the accuracy and adaptability of recommendations. By analyzing historical admission trends and user behavior, the system can evolve into a more intelligent and predictive model. Another potential enhancement is the integration of career guidance features, such as suggesting suitable courses based on student interests and aptitude. The system can also include feedback mechanisms and rating systems to continuously improve recommendation quality.

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